

October Crossword Solutions

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1 2	2 5
3 1	2

ONE DOWN

Note that Kevin and Wenhao have the same parity if and only if $k + w$ is even.

Since $a + k + w = 12$ is even, this is equivalent to Aaron getting an even number of candies. Thus, we have

$$N = \sum_{\substack{a=0 \\ a \text{ even}}}^{12} \binom{12}{a} 2^{12-a} = 2^{12} \sum_{\substack{a=0 \\ a \text{ even}}}^{12} \binom{12}{a} \left(\frac{1}{2}\right)^a = 2^{12} \cdot \frac{\left(1 + \frac{1}{2}\right)^{12} + \left(1 - \frac{1}{2}\right)^{12}}{2} = \frac{3^{12} + 1}{2}.$$

Now, just bash:

$$3^{12} = 531441 \equiv 1441 \pmod{2000}, \quad R = \frac{1441 + 1}{2} \equiv 721 \pmod{1000}.$$

Hence the remainder is $R = 721$, with its final two digits being 21.

THREE ACROSS

We are asked to count the number N of valid circular arrangements of $n = 10$ students, each wearing one of $k = 3$ costumes (Vampire, Ghost, or Zombie), such that no two adjacent students wear the same costume.

It is known (and can be derived from inclusion-exclusion or recurrence relations) that the number of proper colorings of a cycle C_n with k colors is

$$N = (k - 1)^n + (-1)^n (k - 1).$$

Substitute $k = 3$ and $n = 10$:

$$N = (3 - 1)^{10} + (-1)^{10} (3 - 1) = 2^{10} + 2 = 1024 + 2 = 1026.$$

The product of its nonzero digits is just $1 \cdot 2 \cdot 6 =$ 12.

ONE ACROSS

Because the labels $1, 2, \dots, 2025$ are equally distributed among the 5 residue classes $\{0, 1, 2, 3, 4\}$ modulo 5, each residue occurs exactly 405 times. Using the roots of unity filter, the number of subsets with sum congruent to 0 (mod 5) is:

$$\frac{1}{5} \sum_{j=0}^4 \prod_{r=0}^4 (1 + \omega^{jr})^{405},$$

where $\omega = e^{2\pi i/5}$. For $j = 0$, we get:

$$\prod_{r=0}^4 (1 + \omega^{0 \cdot r})^{405} = (1 + 1)^{2025} = 2^{2025}.$$

For $j = 1, 2, 3, 4$, we have:

$$\prod_{r=0}^4 (1 + \omega^{jr}) = \prod_{r=0}^4 (1 + \omega^r) = (x^5 - 1)/(x - 1) \Big|_{x=-1} = (-1)^5 - 1 = 2.$$

Hence each of these four terms contributes 2^{405} . Therefore, the number of pleasing piles is:

$$\frac{1}{5} (2^{2025} + 4 \cdot 2^{405}) = \frac{2^{405}(2^{1620} + 4)}{5}.$$

We can express this in the form:

$$\frac{1}{5} (2^{2025} + 2^{407}).$$

Finally, we compute:

$$\text{Sum of digits of } x = 2 + 0 + 2 + 5 = 9,$$

$$\text{Sum of digits of } y = 4 + 0 + 7 = 11,$$

$$m + (9 + 11) = 5 + 20 = \boxed{25}.$$

TWO DOWN

Each midnight, a given pumpkin explodes iff it and its two neighbors all glow, which occurs with probability $(\frac{1}{2})^3 = \frac{1}{8}$. By linearity of expectation, the expected count multiplies by $1 - \frac{1}{8} = \frac{7}{8}$ each night. Thus,

$$E_n = 64 \left(\frac{7}{8} \right)^n = 2^6 \cdot \frac{7^n}{2^{3n}} = \frac{7^n}{2^{3n-6}}.$$

Since 7^n is odd, the fraction is in lowest terms, and its decimal expansion terminates with exactly $3n - 6$ digits after the decimal point. Given that there are 150 such digits, $3n - 6 = 150 \Rightarrow n = \boxed{52}$, as desired.